

Example: Determine whether the following sequence,

$$\left\{ (-1)^n \frac{\cos(n)}{n^2} \right\}$$

converges or diverges.

$$a_n = \left(-\frac{1}{2}\right)^n \cos(n)$$

$$|a_n| = \frac{|\cos(n)|}{2^n}$$

$$\lim_{n \rightarrow \infty} \frac{|\cos(n)|}{2^n}$$

We can't say much about this yet,

We can use the Squeeze theorem as the function method does not help here

can't say anything about this

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = \lim_{n \rightarrow \infty} -\left(\frac{1}{2}\right)^n = 0$$

Is this because of the absolute thing?

$$\text{So } \lim_{n \rightarrow \infty} \left(-\frac{1}{2}\right)^n \cos(n) = 0$$

$$\left(-\frac{1}{2}\right)^n \leq a_n = \left(-\frac{1}{2}\right)^n \cos(n) \leq \left(\frac{1}{2}\right)^n = b_n$$

Montone Sequences: We say a sequence is *monotone* if it does not oscillate. More specifically, a sequence $\{a_n\}$ is:

- **Monotonically Increasing** if $a_n \leq a_{n+1}$ for all $n = 1, 2, \dots$
 * always getting bigger or staying the same
- **Monotonically Decreasing** if $a_n \geq a_{n+1}$ for all $n = 1, 2, \dots$
 * always getting smaller or staying the same

a function that is constant will be both monotonically increasing and decreasing

Bounded Sequences: We say a sequence $\{a_n\}$ is:

- **Bounded Above** if there is a number M such that $a_n \leq M$ for all $n = 1, 2, \dots$
 * Some number so that every term is less than or equal to that number
- **Bounded Below** if there is a number m such that $a_n \geq m$ for all $n = 1, 2, \dots$
 * Some number so that every term is greater than or equal to that number

If both statements above are true, we say $\{a_n\}$ is a **bounded sequence**.

Example: Consider

Does this sequence converge?

$$\left\{ \frac{n}{n+1} \right\}$$

$$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$$

Bounded above by 1 $\rightarrow$ monotonically increasing $\rightarrow$ prove it bounded below by $\frac{1}{2}$? Or 0?
 * still bounded though

Difference b/w a sequence and a function, functions are continuous

What you want to show: $a_n \leq a_{n+1}$ where $a_n = \frac{n}{n+1}$
 and $a_{n+1} = \frac{n+1}{n+2}$
 $\frac{n}{n+1} \leq \frac{n+1}{n+2}$
 $\frac{n}{n+1} \leq \frac{n+1}{n+2} \xrightarrow{\text{cross}} n(n+2) \leq (n+1)^2 \rightarrow n^2 + 2n \leq n^2 + 2n + 1$

you could also look at $f(x) = \frac{x}{x+1}$ since it's cont. and where f' is positive it's increasing
 $f'(x) = \frac{1}{(x+1)^2}$

We need to prove that it's monotonically increasing this way to show that it holds for all values of n

Theorem: Every bounded monotone sequence converges.

above AND below

monotonically increasing or decreasing
 * even monotone, can't oscillate

Example: Find the limit of the sequence $\{a_n\}$ given by the recurrence relation

$$a_1 = 2 \quad a_{n+1} = \frac{1}{2}(a_n + 6) \quad \text{for all } n \geq 1.$$

$$a_1 = 2, a_2 = 4, a_3 = 5, a_4 = \frac{11}{2}$$

Does this converge? Can we find what it converges to?

Is it bounded? maybe bounded above by 6?

Recursive means we have

to start w $a_1 = 2$ and plug in the rest

$$a_1 = 2 < 6 \quad \text{suppose } a_k \leq 6 \text{ for all } k = 2, \dots, n$$

Still how does this work

We could also say it's bounded above by 6 + that would still be true so you can't infer limit behavior. We just need an upper bound to prove it's bounded but it doesn't have to be exact

It's also bounded below because it will never be negative $\Rightarrow 0 \leq a_n \leq 6$ for all $n \geq 1$

Given a starting point that we establish assume we improve the rest

Is it monotone?

$$a_1 = 2 < a_2 = 4$$

Suppose $a_n \leq a_{n+1}$ for some arbitrary n . Want to show: $a_{n+1} \leq a_{n+2}$

$$a_{n+1} = \frac{1}{2}(a_n + 6) \leq \frac{1}{2}(a_{n+1} + 6) = a_{n+2} \quad \text{So } a_n \leq a_{n+1} \text{ for all } n \geq 1$$

It's bounded and monotone, so it converges. What does it converge to?

We can't take the derivative b/c it's recursively defined

We know it converges so $\lim_{n \rightarrow \infty} a_n = L$

$\lim_{n \rightarrow \infty} a_{n+1}$ will also be L then

$$\lim_{n \rightarrow \infty} a_{n+1} = \frac{1}{2}(\lim_{n \rightarrow \infty} a_n + 6) = L$$

$$L = \frac{1}{2}(L + 6)$$

$$L = 6$$

Exercise: Find the limit of the sequence $\{a_n\}$ given by the recurrence relation

$$a_1 = 1 \quad a_{n+1} = 4 - a_n \quad \text{for all } n \geq 1.$$

$$\{a_n\} = 1, 3, 1, 3, \dots$$

not monotone $\rightarrow$ divergent
or it oscillates

What happens if you change $a_1 = 2$?

$$\{a_n\} = 2, 2, 2, 2, \dots$$

monotone & constant
bounded
when you take limit
of something that never
changes, it approaches that
constant value

$$a_1 = 1 \leq a_2 = 3$$

$$a_2 = 3 \neq a_3 = 1$$